

Automated Lead Generation & Tracking CRM

SYSTEM ARCHITECTURE & SPECIFICATION DOCUMENT

1. Executive Summary & Scope

This document outlines the architectural blueprint and operational specifications for an automated high-efficiency lead acquisition pipeline and Customer Relationship Management (CRM) system. The platform is explicitly optimized for targeting localized business entities within **Kenya** by leveraging public data from the Google Places API ecosystem.

The system achieves architectural balance through a **Human-in-the-Loop** operational paradigm. While data collection, geographic validation, formatting, and history tracking are fully automated via server-side routines, message delivery is executed client-side via native user actions. This design guarantees complete protection against platform-level automated spam detection mechanisms and complies with core structural data constraints.

2. Core System Architecture

The application follows a decoupled three-tier architecture structured to separate non-blocking automated background workflows from interactive real-time management views.

2.1 The Ingestion Tier (Background Daemon)

An autonomous, scheduling engine responsible for structural polling. It wakes up at predefined cron intervals, constructs optimized geographic payloads, targets specific business categories, filters incoming responses based on deterministic spatial fields, and commits fresh records into persistence.

2.2 The Persistence Tier (Data Core)

The central source of truth enforcing uniqueness constraints. It rejects multi-pass data collection duplicates, manages state transitions for records across their lifecycle, and stores historical operational logs.

2.3 The Presentation Tier (Operator Dashboard)

A lightweight dashboard configured to read active pipeline views. It translates persistence states into operational UI layouts and dynamically generates standard hyperlink endpoints to safely trigger outreach pipelines.

3. Database Design & Data Model

To successfully deduplicate entities across infinite collection sweeps and preserve strict operational state tracking, the core entity model enforces the following logical parameters:

FIELD NAME	DATA TYPE	INDEX TYPE	FUNCTIONAL PURPOSE
id	ObjectId / UUID	Primary Key	System-wide unique internal primary reference identifier.
placeId	String	Unique Index	Immutable location string identifier returned by the Google Places API. Prevents identical mapping.
name	String	Standard	The formal trading name of the commercial listing.
address	String	Standard	Full text physical location and street routing layout metadata.
whatsappNumber	String	Unique Index	Strictly cleaned E.164 international standard numeric string (e.g., 2547XXXXXXX). Ensures absolute outreach deduplication.
status	Enum String	State Index	Restricted to explicit states: Pending, Contacted, or Archived. Defines visibility.
fetchedAt	DateTime	Standard	Timestamp tracking exactly when the entry was processed into the system.
contactedAt	DateTime (Nullable)	Standard	Records the exact moment the operator initiated the external communication link.

4. Data Ingestion & Geographic Safeguards

To isolate queries safely within Kenya's domestic borders and maintain rigid cost control over APIs, the ingestion engine deploys three concurrent safety parameters:

4.1 Strict Polygon Bounding (API Restriction)

Every search payload forces spatial parameters via a `locationRestriction` rectangle mapping the absolute outermost coordinate limits of Kenya. This guarantees that ambiguous search terms never pull global results:

- **Southern Boundary Limit:** Latitude -4.71
- **Western Boundary Limit:** Longitude 33.84
- **Northern Boundary Limit:** Latitude 4.63
- **Eastern Boundary Limit:** Longitude 41.92

4.2 Multi-Stage Fail-Safe Filtering

Before an item is parsed into memory, it passes through an isolated validation array:

1. **String Normalization:** The physical address field is stripped and validated to confirm it contains the keyword "Kenya". If absent, the data item is dropped.

- 2. **Number Transposition:** Local phone strings (such as 07 . . . or 01 . . .) are parsed, stripped of special characters or spacing, and programmatically prefixed with the 254 international country identifier.

5. Intelligent Rate Limiting & Deduplication Logic

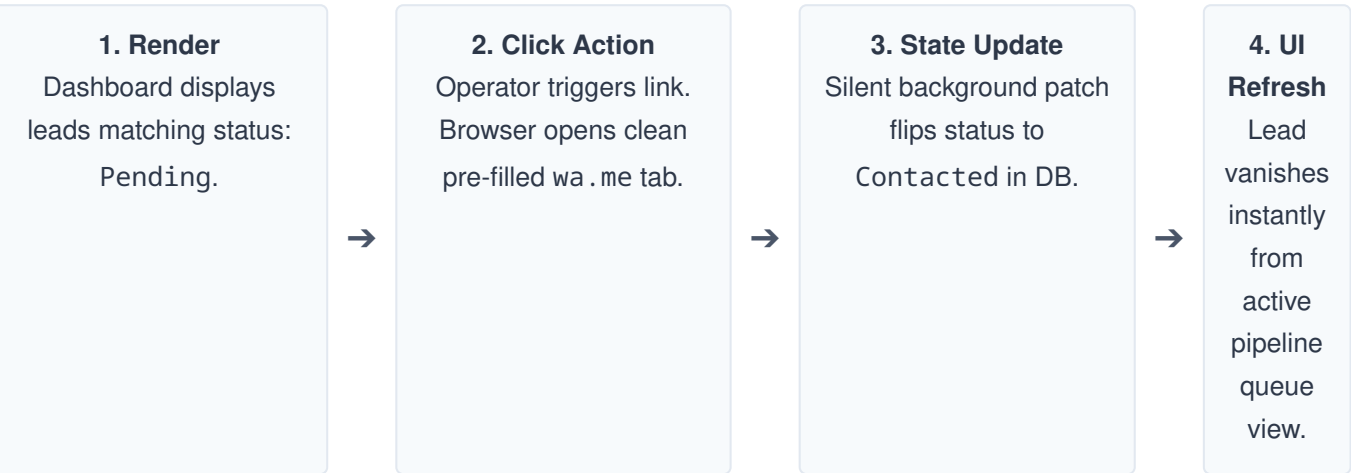
To ensure the system collects exactly **10 new, uncontacted businesses per day** without burning through API limits on previously stored listings, the daemon follows a specific quota-filling state machine:

The Quota Ingestion Flow Algorithm

- Instead of requesting a static count of 10 items from Google (which could all be items you already messaged last week), the loop follows this architecture:
1. The scheduler executes an API pull requesting a wide localized page batch (e.g., 30 listings).
 2. An internal transaction counter is initialized: `savedCount = 0`.
 3. The system iterates through the array sequentially. For each item:
 - It queries the database for an existing match using the unique `whatsappNumber` or `placeId`.
 - If a match exists, the item is instantly discarded from memory.
 - If no match exists, the engine inserts the record into the database with a state of `Pending`, and increments the counter: `savedCount++`.
 4. The moment `savedCount === 10`, the iteration loop breaks instantly, truncates all remaining unparsed entities, and safely places the background service back into idle mode.

6. The Interactive Human-in-the-Loop Pipeline

The system purposefully avoids automated API messaging scripts to maintain complete protection against account flags. Instead, the workflow relies on dynamic browser state management:



6.1 Status Pipeline States

- **Pending Queue:** The standard workspace view showing unaddressed entries.

- **Contacted Ledger:** A read-only audit panel displaying entities already approached, complete with temporal tracking for analytical reviews.
- **Archived Vault:** Houses invalid entries or locations flagged manually as irrelevant, keeping them permanently out of sight while preserving their unique key states so they are never re-scraped in future sweeps.